

# George David Tsitlauri

## AI & Systems Research Engineer

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**Specialization:** Data Management & AI · **Lab:** CACS Lab (Computer Architecture, Compilers & System Security)

## EDUCATION

**MSc in Artificial Intelligence & Applications** · University of Thessaly, Greece 2026–2027

Thesis: *TELEMACHUS: Certified Information Acquisition and Escalation in AI Agents*

Supervisor: Prof. G. Dimitriou

**BSc in Informatics & Telecommunications** · University of Thessaly, Greece 2021–2026

Thesis: *CryptoCPU: Design and Implementation of an Encrypted Instruction Set Processor on FPGA*

Supervisor: Prof. G. Dimitriou

## RESEARCH

**Crypto3DStackCPU** · Secure processor with 3D-stacked memory

A processor that runs only encrypted, signed programs on top of a four-layer 3D-stacked memory. Every transfer between the memory layers is cryptographically signed, so tampering, replayed data and physical fault attacks are detected, and the chip locks itself and erases its keys when it is attacked.

Supervisor: Prof. G. Dimitriou · Preliminary PhD research

**CUDA-zkML** · Verifiable neural-network inference on GPUs

A system that lets a service prove that an answer really came from the neural network it claims, without revealing the model or the user's data. The short cryptographic proofs are generated on a graphics card and can be checked in milliseconds, even by a smart contract.

Supervisor: Prof. G. Dimitriou · Preliminary PhD research

## ACADEMIC PROJECTS

**NEXUS** · compiler for a small C-like language with register allocation and SSA, CPU simulators from single-cycle to out-of-order (Tomasulo), multicore and GPU models, and a pipelined MIPS processor in Verilog

**CCC IR Optimizer** · optimizer in C for the CCC high-level-synthesis toolchain: region-based and iterative data-flow analysis, constant propagation, loop-invariant code motion, common-subexpression and dead-code elimination

**Parallel AES** · AES-128 encryption on NVIDIA GPUs in CUDA C++ (ECB and CTR modes), with one GPU thread per 16-byte block, verified against FIPS-197 and OpenSSL and reaching 12–15 GB/s

**GPBFT** · Byzantine fault-tolerant consensus protocol that groups nodes and aggregates their votes to reduce communication, implemented in Python and evaluated against PBFT in the SymbChainSim blockchain simulator

**Kubernetes Autoscaler** · cloud-native weather-forecast service with a custom autoscaler that adds or removes pods from the measured request rate under oscillating load

## WEBSITES

**Skyrianos** · website about the island of Skyros with information for visitors

**OstrichZooPark** · website for an ostrich zoo park with visitor information

**Pavlidi Home** · e-commerce website with a product catalogue and online ordering

## SKILLS

|                    |                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python, C, C++, Java, SQL, JavaScript, TypeScript, Assembly, VHDL, Verilog                |
| <b>AI / Data</b>   | PyTorch, TensorFlow, XGBoost, Apache Spark, Apache Hadoop                                 |
| <b>Hardware</b>    | CUDA, OpenMP, TensorRT, Xilinx Vivado, Vitis HLS                                          |
| <b>Web</b>         | HTML5, CSS3, React, Next.js, Node.js, Tailwind CSS                                        |
| <b>Dev</b>         | Git, CI/CD, Linux, Bash, CMake, Docker, Kubernetes, Wireshark, PostgreSQL, Jupyter, LaTeX |

## LANGUAGES

|                |                               |
|----------------|-------------------------------|
| <b>Greek</b>   | Native                        |
| <b>Russian</b> | Native                        |
| <b>English</b> | ECCE – University of Michigan |